

18.S097 Introduction to Proofs

IAP 2015

Example Sheet 1

Example 1. Show that “ $p \Rightarrow q$ ” is equivalent to “ $\text{not}(q) \Rightarrow \text{not}(p)$ ” for any two propositions p and q .

Hint: Recall that the validity of $p \Rightarrow q$ is defined by the validity of the statement “ $\text{not}(p)$ or q ”.

Proof. Let p and q be given propositions. Suppose first that $p \Rightarrow q$ holds, i.e. that the statement “ $\text{not}(p)$ or q ” is true. We want to show that

$$\text{“not}(\text{not}(q)) \text{ or } \text{not}(p)\text{”} \tag{1}$$

is true, which, by definition, is equivalent to $\text{not}(q) \Rightarrow \text{not}(p)$ (thus implying the first half of the desired equivalence). The validity of the statement (1) follows by observing that, for every proposition r , the proposition “ $\text{not}(\text{not}(r))$ ” is equivalent to r , and that, for every proposition r and s , the propositions “ r or s ” and “ s or r ” are equivalent (each of these equivalences can be verified by examining the relevant truth tables).

Conversely, suppose that $\text{not}(q) \Rightarrow \text{not}(p)$ holds. By definition, this means that either $\text{not}(\text{not}(q))$ or $\text{not}(p)$ are true. As in the previous argument, this pair of conditions is equivalent to q or $\text{not}(p)$, which is equivalent to $p \Rightarrow q$, as desired. $\square$

Example 2. Let X be a given set and let A, B be two arbitrary subsets of X . Show that

$$A \cap B \subset A.$$

Proof. Let $x \in A \cap B$ be given. By the definition of the intersection of sets, we then have the statement “ $x \in A$ and $x \in B$ ” (so that in particular $x \in A$ holds). Since $x \in A \cap B$ was arbitrary, this shows the desired set inclusion. $\square$

Example 3. Let X, A and B be as above. Show that

$$(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B).$$

Proof. We show the desired equality in two steps: by showing (i) $(A \setminus B) \cup (B \setminus A) \subset (A \cup B) \setminus (A \cap B)$, and (ii) $(A \cup B) \setminus (A \cap B) \subset (A \setminus B) \cup (B \setminus A)$.

(i): Let $x \in (A \setminus B) \cup (B \setminus A)$ be given. Then either $x \in A \setminus B$ or $x \in B \setminus A$. Suppose first that $x \in A \setminus B$. We then have $x \in A \subset A \cup B$. In order to show $x \in (A \cup B) \setminus (A \cap B)$, it remains to show that $x \notin A \cap B$. Suppose for contradiction that $x \in A \cap B$ holds. This in particular implies $x \in B$, contradicting $x \in A \setminus B$ (which holds by assumption). It follows that our original assumption $x \in A \cap B$ is false, so that $x \notin A \cap B$ holds as desired. We have therefore shown $x \in (A \cup B) \setminus (A \cap B)$ in this case.

Alternatively, suppose that $x \in B \setminus A$, so that $x \in B$ and $x \notin A$. It now follows from $B \subset A \cup B$ that $x \in A \cup B$, while $x \notin A$ implies $x \notin A \cap B$ (note that $A \cap B \subset A$, so that if $x \in A \cap B$ held, we would have $x \in A$)¹. Thus $x \in (A \cup B) \setminus (A \cap B)$ holds in this case as well.

¹This is really just a rephrasing of the argument in the previous paragraph.

Since x was arbitrary, we have shown the desired inclusion.

(ii): Let $x \in (A \cup B) \setminus (A \cap B)$ be given. We then have $x \in A \cup B$ and $x \notin A \cap B$. We split into two cases:

Case 1: $x \in A$.

In this case, $x \notin A \cap B$ implies $x \notin B$ (this is the contrapositive of the fact that for $x \in A$ one has the implication $x \in B$ implies $x \in A \cap B$). We therefore have $x \in A \setminus B \subset (A \setminus B) \cup (B \setminus A)$.

Case 2: $x \notin A$.

In this case, the condition $x \in A \cup B$ implies $x \in B$. On the other hand, $x \in B$ and $x \notin A \cap B$ together imply $x \notin A$ (otherwise, we would have $\square$

Remark 1. Some duplication in the previous proof can be avoided by noticing that the two cases in each of (i) and (ii) are symmetric in A and B . In particular, the proof of the second case in each of (i) and (ii) can be obtained from the first case by interchanging the roles of A and B . This can be made precise by formulating an intermediary claim; for instance note that in (i) both cases in the proof can be handled by showing the claim: For every $C, D \subset X$, one has $C \setminus D \subset (C \cup D) \setminus (C \cap D)$ (which can be established by proceeding as in the argument for either case), and taking $(C, D) = (A, B)$ followed by $(C, D) = (B, A)$.